Block 5: Drug price elasticities. The calibration targets neither the participation nor the intensive elasticity of drug demand. The model's participation elasticity is -1.32, somewhat larger in absolute value than the estimates of about -0.3 to -0.9 in Saffer and Chaloupka (1999) and Grossman and Chaloupka (1998), and its intensive elasticity, over all users, is -0.63, in the middle of the estimates of about -0.3 to -0.8 in Grossman and Chaloupka (1998) and Olmstead et al. (2015). These elasticities are the response of drug use to its price, which is separate from the response to Treatment First itself. The latter, with the drug price held fixed, is matched to the evidence on impact when the detection rate is calibrated (Section 6). Given that, a larger price response can only work against Treatment First: as Treatment First lowers the drug price, more households return to using, and the more elastic participation is, the more of them do. A welfare gain from Treatment First is therefore conservative in this respect.

Block 6: Transmission. A parent's SUD and homelessness also carry over into its child's adult life, in the model as in the data. A child of a parent with a SUD has 1.46 times the odds of acquiring one in the model, within the range of 1.44 to 2.51 in Mellentin et al. (2016) and Yoon et al. (2013). An adult who is homeless has 6.23 times the odds of having had a homeless parent, above the 1.8 to 2.4 of Bassuk et al. (1997).33

Transmission in the model is therefore at the low end of the evidence on the SUD, the margin Treatment First acts on, and above it on homelessness, the margin Housing First acts on. With the penalty switched off, in the economy without the externalities of Section 6.2, the SUD odds ratio is 1.02, so the transmission runs through δd. The value of δd that would put the model in the middle of the range is above 0.24, so the calibrated value is likely at the low end of what the evidence supports. The model loads the externality on children towards Housing First more than Treatment First, so if Treatment First still raises welfare in the experiments below, that result is conservative.

Furthermore, in the model, 5.82 percent of children live with a parent who has a SUD; in NSDUH 2024, 4.6 percent of children whose parents are aged 26 or older live with a parent who has a moderate or severe drug use disorder. The externalities on the child in the model therefore reach a share of children similar to that in the data.

Block 7: Losing a voucher. Enrolling in Housing First was estimated to lower the three-year mortality of homeless veterans with mental illness by 4.6 percentage points (Costantini, 2025). In the model, with the drug price and πV held, I compare households with SMI whom Treatment First's revocation puts on the street with the same households had they kept their voucher: losing it raises three-year mortality by about 3.6 percentage points, the same order of magnitude. This matters for the experiments because the households whose voucher Treatment First revokes bear its most direct cost, and the model's estimate of what losing a voucher does to their survival is of the size Costantini measured.

32The rule sets the tenant's contribution at thirty percent of its income, but the voucher pays only up to the fair market rent, so a holder that rents a dwelling above it pays the difference itself, which raises its share above thirty percent, as it does in HUD's data.

33Bassuk et al. (1997) compare 220 homeless mothers with 216 low-income housed mothers, so both groups are poor and the odds ratios, 1.8 for a homeless female caretaker and 2.4 for a male one, separate the transmission of homelessness from the poverty around

it; a ratio against the general population would credit the housing penalty with everything else that differs between poor children and the rest.

Table 4: Untargeted moments

Moment Data Model Source

Block 1: Homelessness Share of homeless by age, 25–34 to 65 + a 26.0, 27.2, 21.0, 18.4, 7.5 26.4, 37.9, 24.2, 10.7, 0.8 HUD PiT 2024 Employed share of the homeless 46 21.4 Meyer et al. (2024) Income of the homeless, over average 19.1 6.5 Meyer et al. (2024)

Block 2: Who uses and who has a SUD Share of users by age, 26–34 to 65 + 45, 19, 14, 12, 10 67, 16, 10, 5, 2 NSDUH 2023 Mean age of users, years 39.6 34.6 NSDUH 2023 Unemployment given a SUD 16.2 14.5 NSDUH 2024 Drug use with SMI over without, ratio 2.44 2.64 NSDUH 2024 Drug deaths as a share of deaths, 25–44 20.0 14.5 CDC WONDER

Block 3: Housing and vouchers Renters paying over 30% of income 43.2 58.7 SCF 2022 Rent over income, voucher holders 31.2 39.0 HUD PSH 2023 Voucher holders living on wages 28.0 39.2 HUD PSH 2023

Block 4: Income and wealth distributions, shares of the total Income, bottom 1, 5, 10, 20% 0.01, 0.31, 0.92, 2.71 0.01, 0.19, 1.00, 3.40 SCF 2022 Net worth, bottom 1, 5, 10, 20, 50% -0.16, -0.28, -0.29, -0.23, 2.2 0.00, 0.00, 0.00, 0.27, 6.3 SCF 2022

Block 5: Drug price elasticities Participation (extensive margin) -0.3 to -0.9 -1.32 Saffer and Chaloupka (1999), Grossman and Chaloupka (1998) Intensive margin (users) -0.3 to -0.8 -0.63 Grossman and Chaloupka (1998), Olmstead et al. (2015)

Block 6: Transmission Child of a SUD parent acquires a SUD, odds ratio 1.44, 2.51 1.46 Mellentin et al. (2016), Yoon et al. (2013) Homeless adult had a homeless parent, odds ratio 1.8 to 2.4 6.23 Bassuk et al. (1997) Children living with a parent with a SUD, % 4.6 5.8 NSDUH 2024

Block 7: Losing a voucher Three-year mortality, households with SMI put on the street, ppb 4.6 3.6 Costantini (2025)

Block 8: Treatment under a condition Abstinence, contingent over unconditional housing, ratioc 1.26 1.40 Milby et al. (2005)

Notes: percent, except the elasticities and the odds ratios of the transmission block, the ratio in Block 8, the mean age (years) and the SMI ratio in Block 2, and percentage points in Block 7; the model is the Housing First benchmark, except in Blocks 7 and 8, which apply Treatment First to it (notes b and c). aThe share aged 25 to 34 is the target of T0 in Table 3, so of this row only the other four bands are untargeted. bData: Costantini's instrumental-variables estimate of how much enrolling in the Housing First programme of the Department of Veterans Affairs lowers the three-year mortality of homeless veterans with mental illness. Model: the three-year mortality gap between the households with SMI whom Treatment First's revocation forces into homelessness and the same households had they kept their voucher, with the drug price and πV held. cData: consecutive weeks of abstinence in the arm housed on condition of abstinence over the arm housed without it,

homeless adults with a cocaine disorder in day treatment. Model: the recovery hazard of voucher holders with a SUD over that of housed households with a SUD and no voucher, in the first year of Treatment First.

Block 8: Treatment under a condition. Milby et al. (2005) randomised abstinence-contingent housing against housing without conditioning, among homeless adults with a cocaine disorder in treatment: the contingent arm reached 8.32 consecutive weeks of abstinence against 6.60, a ratio of 1.26. The model's counterpart is the recovery hazard of voucher holders, whose voucher is conditional under Treatment First, against housed households without a voucher, both with a SUD: 1.40 times in the first year of the condition, the horizon of Milby's trial, close to its ratio. This matters for the experiments because treatment is how Treatment First reaches the households that already have a SUD, and the model's response over that horizon is of the size the trial measured.

6 Experiments and results

Every experiment replaces Housing First, the benchmark, with Treatment First, the rule of Section 4.9, and keeps the programme rationed and the tax system as it is in the benchmark. Two objects can respond to the reform: the drug price, by clearing the drug market, and the probability of drawing a voucher πV , by holding net government revenue at its benchmark level $\bar{R}$, so that what the screen saves is spent on vouchers paid to others. The experiments differ in which of the two respond and in whether the externalities on children are present. There are three experiments, and each answers its own question. The first two compare steady states (Section 6.2), each in its own half of Table 5, and the third considers the transition (Section 6.4).

The first experiment asks whether moral hazard alone justifies conditionality. In economics, moral hazard is defined as the effect insurance has on the incentive to avoid the loss it covers: when part of the cost of a loss falls on someone else, the insured does less to avoid it, which makes the loss more likely and raises what it costs whoever shares in it (Arrow, 1963; Pauly, 1968). Housing First insures a household against losing its home when drug use leads to a SUD, the SUD to unemployment and unemployment to homelessness. Since the assistance does not depend on use, part of what using costs falls on the government's budget, and through it on others who fund the programme. This is the case critics of Housing First normally make.

To test whether moral hazard justifies Treatment First, I remove the children from welfare and from the parent's preferences, and make their productivity independent of what the parent did. A household's drug use then harms no one but itself and the budget, so any gain comes from correcting moral hazard rather than from correcting the externalities on the children. I then re-solve the model under Housing First with these switches, keeping every calibrated parameter fixed, which gives the benchmark without externalities. I introduce Treatment First into it in two ways, both holding Housing First's drug price fixed. First, πV is also held at its Housing First level, so that only the screen acts, which shows what deterring users does to the users themselves. Second, πV clears the budget, so the savings are spent on vouchers for others; this run measures the gain from correcting moral hazard's cost to the budget that critics of Housing First have in mind. If Treatment First lowers welfare in these cases, moral hazard alone is no reason to make the assistance conditional. For completeness, I also show the case in which the drug price clears together

with πV .